~Deepak Patil - 31359

Scala program using Apache Spark

Apache Spark installation

1. Update your system

```
sudo apt update && sudo apt upgrade -y
```

2. Install Java (OpenJDK 11 or 17)

```
sudo apt install openjdk-11-jdk -y
java -version
```

3. Install Scala

```
sudo apt install scala -y
scala -version
```

4. Download and Install Spark

```
wget https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
tar -xvf spark-3.5.0-bin-hadoop3.tgz
sudo mv spark-3.5.0-bin-hadoop3 /opt/spark
```

5. Configure Environment Variables

```
echo 'export SPARK_HOME=/opt/spark' >> ~/.bashrc
echo 'export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin' >> ~/.bashrc
source ~/.bashrc
```

Writing a Simple Spark Program - Word Count
Create a file named WordCount.scala

```
nano WordCount.scala
```

(paste below code)

```
import org.apache.spark.sql.SparkSession
object WordCount {
  def main(args: Array[String]): Unit = {
    val spark = SparkSession.builder()
      .appName("Simple Word Count")
      .master("local[*]") // Uses all available cores on your machine
      .getOrCreate()
    val data = Seq("Spark is fast", "Spark is fun", "Scala is powerful")
    val rdd = spark.sparkContext.parallelize(data)

    val counts = rdd
      .flatMap(line => line.split(" "))
      .map(word => (word, 1))
      .reduceByKey(_ + _)

    counts.collect().foreach(println)
```

```
    spark.stop()
  }
}
```

Execution

1. Install SBT

```
echo "deb https://repo.scala-sbt.org/scalasbt/debian all main" | sudo tee
/etc/apt/sources.list.d/sbt.list
echo "deb https://repo.scala-sbt.org/scalasbt/debian /" | sudo tee
/etc/apt/sources.list.d/sbt_old.list
curl -sL
"https://keyserver.ubuntu.com/pks/lookup?op=get&search=0x2EE0EA64E40A89B84B2DF73
499E82A75642AC823" | sudo apt-key add
sudo apt-get update
sudo apt-get install sbt
```

2. Create the Project Structure

```
mkdir -p simple-spark-project/src/main/scala
cd simple-spark-project
```

Move your WordCount.scala file into the new folder:

```
mv ../WordCount.scala src/main/scala/
```

3. Create the Build Definition (build.sbt)

>> nano build.sbt (paste below text)

```
name := "SimpleSparkProject"
version := "1.0"
scalaVersion := "2.12.18" // Match this to your installed Scala version

libraryDependencies += "org.apache.spark" %% "spark-sql" % "3.5.0"
```

>> ctrl+O - enter - ctrl+X

4. Compile and Package

```
sbt package
```

Once finished, you will see a message pointing to your JAR file, usually located at:
target/scala-2.12/simplesparkproject_2.12-1.0.jar

5. Execute with spark-submit

```
spark-submit \
  --class "WordCount" \
  --master "local[*]" \
  target/scala-2.12/simplesparkproject_2.12-1.0.jar
```

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